

4.3 Topic 7: Run for your life

Students will be assessed on their ability to:

- 1 Demonstrate knowledge and understanding of the *How Science Works* areas listed in the table on page 12 of this specification.
- 2 Describe the structure of a muscle fibre and explain the structural and physiological differences between fast and slow twitch muscle fibres.
- 3 Explain the contraction of skeletal muscle in terms of the sliding filament theory, including the role of actin, myosin, troponin, tropomyosin, calcium ions (Ca^{2+}), ATP and ATPase.
- 4 Recall the way in which muscles, tendons, the skeleton and ligaments interact to enable movement, including antagonistic muscle pairs, extensors and flexors.
- 5 Describe the overall reaction of aerobic respiration as splitting of the respiratory substrate (e.g. glucose) to release carbon dioxide as a waste product and reuniting of hydrogen with atmospheric oxygen with the release of a large amount of energy.
- 6 **Describe how to investigate rate of respiration practically.**
- 7 Recall how phosphorylation of ADP requires energy and how hydrolysis of ATP provides an accessible supply of energy for biological processes.
- 8 Describe the roles of glycolysis in aerobic and anaerobic respiration, including the phosphorylation of hexoses, the production of ATP, reduced coenzyme and pyruvate acid (details of intermediate stages and compounds are not required).
- 9 Describe the role of the Krebs cycle in the complete oxidation of glucose and formation of carbon dioxide (CO_2), ATP, reduced NAD and reduced FAD (names of other compounds are not required) and that respiration is a many-stepped process with each step controlled and catalysed by a specific intracellular enzyme.
- 10 Describe the synthesis of ATP by oxidative phosphorylation associated with the electron transport chain in mitochondria, including the role of chemiosmosis and ATPase.